Application Layer Definitions

2.1 Principles of Network Applications

Client-Server Architecture A model where an always-on host (the *server*) services requests from many other hosts (*clients*). Clients do not communicate directly with each other.
Peer-to-Peer (P2P) Architecture A model with no dedicated server. Hosts (*peers*) communicate directly, acting as both clients and servers. Known for its *self-scalability.*
Process A program running on a host. A network application consists of communicating processes on different hosts.
Socket The Application Programming Interface (API) that acts as a "door" between an application process and the transport layer. An application sends/receives all network messages through a socket.
IP Address & Port Number The complete identifier for a network process. The IP address identifies the host, and the port number identifies the specific process on that host.
TCP (Transmission Control Protocol) A reliable, connection-oriented transport protocol. It provides in-order data delivery, flow control, and congestion control.
UDP (User Datagram Protocol) An unreliable, connectionless transport protocol. It offers minimal, "best-effort" service with no guarantees.
TLS (Transport Layer Security) An application-layer enhancement over TCP that provides encryption, data integrity, and authentication for secure communication (forms HTTPS).

2.2 Web and HTTP

HTTP (Hypertext Transfer Protocol) The web's application-layer protocol. It is a *pull protocol* (clients pull data from servers) and is **stateless** (server maintains no info about past requests).
Non-persistent HTTP One TCP connection is used for each object. Inefficient, requiring 2 RTTs of overhead per object.
Persistent HTTP Multiple objects are transferred over a single TCP connection. The modern default, much more efficient.
RTT (Round-Trip Time) The time for a small packet to travel from client to server and back.
Cookies A mechanism for websites to track user state. The server sends a unique ID to the client, which the client returns in subsequent requests.
- **First-Party Cookie:** Cookie from the domain the user is directly visiting.
- **Third-Party Cookie:** Cookie from a different domain (e.g., an ad network), used to track users across multiple websites.

Web Cache (Proxy Server) A network entity that stores copies of recently requested web objects to satisfy future requests locally, reducing response time and traffic.
Conditional GET An HTTP request that includes an 'If-Modified-Since' header. Allows a cache to verify if its stored object is up-to-date without re-downloading it if it's unchanged (server returns '304 Not Modified').
HTTP/2 Mitigates Head-of-Line (HOL) blocking by dividing objects into frames and interleaving them on a single TCP connection.
HTTP/3 Eliminates transport-layer HOL blocking by using the QUIC protocol over UDP instead of TCP.

2.3 Email Protocols

User Agent (UA) Client software for email (e.g., Outlook).
Mail Server An always-on host that stores incoming emails in user **mailboxes** and outgoing emails in a **message queue**.
SMTP (Simple Mail Transfer Protocol) The protocol for **pushing** email from a client to a server, and between servers. Uses TCP port 25.
Mail Access Protocol A protocol for **pulling** email from a server to a user agent.
- **IMAP (Internet Mail Access Protocol):** Modern protocol where messages are managed on the server, allowing synchronized access from multiple clients.
- **POP (Post Office Protocol):** Older protocol that typically downloads messages to a single client and deletes them from the server.

2.4 The Domain Name System (DNS)

DNS A distributed, hierarchical database that translates human-readable hostnames into IP addresses. Implemented at the application layer, typically using UDP.
DNS Hierarchy • **Root DNS Servers:** Top of the hierarchy; direct queries to TLD servers.
- **Top-Level Domain (TLD) Servers:** Manage domains like '.com', '.org', etc.
- **Authoritative DNS Servers:** Hold the definitive hostname-to-IP mappings for a specific domain.

Local DNS Server A server (e.g., at an ISP) that acts as a proxy, making queries into the DNS hierarchy on behalf of hosts.
Iterated Query DNS resolution where the local server queries each level of the hierarchy directly.

Recursive Query DNS resolution where a server takes on the burden of resolving a query on behalf of another server.
DNS Caching & TTL DNS servers cache query results to improve speed. The **Time-to-Live (TTL)** value in a record dictates how long it can be cached.
Resource Record (RR) A DNS database entry. Types include **A** (address), **NS** (name server), **CNAME** (alias), and **MX** (mail exchange).

2.5 & 2.6 P2P, Video, and CDNs

BitTorrent A P2P protocol for file distribution where a file is broken into **chunks**.
Tracker A server that tracks the peers participating in a **torrent** (the group of peers for a file).
Tit-for-Tat The incentive mechanism where a peer uploads fastest to the peers that upload fastest to it, encouraging cooperation.
DASH (Dynamic Adaptive Streaming over HTTP) The dominant video streaming standard. The client detects its available bandwidth and requests video chunks of a corresponding quality level from the server, using a **Manifest File** that lists all available chunks and qualities.
Playout Buffer A client-side buffer used to absorb network **jitter** (variability in packet delay) to ensure smooth video playout.
CDN (Content Distribution Network) A geographically distributed network of servers that stores copies of content to bring it closer to end-users, improving performance and scalability for services like Netflix.